ITA Classroom Activity 6

Date: 08.08.2025 (Friday)

Hour: 1 Hour

You have:

- Ground truth (P):
  - Apple = 0.8
  - Banana = 0.1
  - Orange = 0.1
- Model Prediction (Q):
  - Apple = 0.6
  - Banana = 0.3
  - Orange = 0.1

Compute the following:

$$KL(P||Q) = \sum P(x) \cdot \log_2 \left(\frac{P(x)}{Q(x)} \right)$$

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Ground Truth P:

- Apple = 0.8
- Banana = 0.1
- Orange = 0.1

Model Prediction Q:

- Apple = 0.6
- Banana = 0.3
- Orange = 0.1

Formula:

$$KL(P \parallel Q) = \sum_x P(x) \cdot \log_2 \frac{P(x)}{Q(x)}$$

Step 1. Term by term calculation

(i) Apple

$$P = 0.8,\ Q = 0.6$$

$$\frac{P}{Q} = \frac{0.8}{0.6} = 1.3333$$

$$\log_2(1.3333) = 0.415$$

$$0.8 \times 0.415 = 0.332$$

(ii) Banana

$$P = 0.1,\ Q = 0.3$$

(ii) Banana

$$P = 0.1,\ Q = 0.3$$

$$\frac{P}{Q} = \frac{0.1}{0.3} = 0.3333$$

$$\log_2(0.3333) = -1.585$$

$$0.1 \times (-1.585) = -0.159$$

(iii) Orange

$$P = 0.1,\ Q = 0.1$$

$$\frac{P}{Q} = 1$$

$$\log_2(1) = 0$$

$$0.1 \times 0 = 0$$

Step 2. Sum them up

$$KL(P \parallel Q) = 0.332 - 0.159 + 0 = 0.173 \text{ bits}$$

✅ **Final Answer:**

$$KL(P \parallel Q) \approx 0.173 \text{ bits}$$

🔑 **Interpretation:**

- The KL divergence is small but not zero → meaning the model Q is close to the true distribution P, but not perfect.
- Most of the divergence comes from **underestimating Apple (0.6 vs 0.8)** and **overestimating Banana (0.3 vs 0.1).**